THE DIMENSION INEQUALITY FOR A DOMINANT LOCALLY FINITE TYPE MORPHISM

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The dimension formula for the local rings at a point and its image. Lemma \ref{lemma-dimension-formula}, cited at proof lines 13766-13772. Gives the middle inequality.
- A chain of specialisations lies in the spectrum of the local ring at its final point. Schemes, Lemma \ref{schemes-lemma-specialize-points}, cited at proof lines 13764-13766. Bounds the length of the chain by the local dimension.

Source reference. Original lemma 02JX: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3352–3352. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 01W1: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2466–2466. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01J7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2004–2004. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02JU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3349–3349. Navigation pointer to a statement the

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Source reference. Original lemma 02J6: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3325–3325. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 01TB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2368–2368. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 02JB: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3330–3330. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 0066: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 234–234. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2. *Let $f : X \to Y$ be a morphism of schemes. Assume that*

(1) *Y is locally Noetherian,*
(2) *X and Y are integral schemes,*
(3) *f is dominant, and*
(4) *f is locally of finite type.*

Then we have

$$\dim(X) \leq \dim(Y) + trdeg_{R(Y)}R(X).$$

If f is closed[1] *then equality holds.*

Proof. Let $f : X \to Y$ be as in the lemma. Let $\xi_0 \rightsquigarrow \xi_1 \rightsquigarrow \ldots \rightsquigarrow \xi_e$ be a sequence of specializations in X. Set $x = \xi_e$ and $y = f(x)$. Observe that $e \leq \dim(\mathcal{O}_{X,x})$ as the given specializations occur in the spectrum of $\mathcal{O}_{X,x}$, see Schemes, Lemma 01J7. By the dimension formula, Lemma 02JU, we see that

$$\begin{aligned} e &\leq \dim(\mathcal{O}_{X,x}) \\ &\leq \dim(\mathcal{O}_{Y,y}) + \mathrm{trdeg}_{R(Y)}R(X) - \mathrm{trdeg}_{\kappa(y)}\kappa(x) \\ &\leq \dim(\mathcal{O}_{Y,y}) + \mathrm{trdeg}_{R(Y)}R(X) \end{aligned}$$

Hence we conclude that $e \leq \dim(Y) + \mathrm{trdeg}_{R(Y)}R(X)$ as desired.

Next, assume f is also closed. Say $\overline{\xi}_0 \rightsquigarrow \overline{\xi}_1 \rightsquigarrow \ldots \rightsquigarrow \overline{\xi}_d$ is a sequence of specializations in Y. We want to show that $\dim(X) \geq d + r$. We may assume that $\overline{\xi}_0 = \eta$ is the generic point of Y. The generic fibre X_η is a scheme locally of finite type over $\kappa(\eta) = R(Y)$. It is nonempty as f is dominant. Hence by Lemma 02J6 it is a Jacobson scheme. Thus by Lemma 01TB we can find a closed point $\xi_0 \in X_\eta$ and the extension $\kappa(\eta) \subset \kappa(\xi_0)$ is a finite extension. Note that $\mathcal{O}_{X,\xi_0} = \mathcal{O}_{X_\eta,\xi_0}$ because η is the generic point of Y. Hence we see that $\dim(\mathcal{O}_{X,\xi_0}) = r$ by Lemma 02JU applied to the scheme X_η over the universally catenary scheme $\mathrm{Spec}(\kappa(\eta))$ (see Lemma 02JB) and the point ξ_0. This means that we can find $\xi_{-r} \rightsquigarrow \ldots \rightsquigarrow \xi_{-1} \rightsquigarrow \xi_0$ in X. On the other hand, as f is closed specializations lift along f, see Topology, Lemma

[1] For example if f is proper, see Definition 01W1.

0066. Thus, as ξ_0 lies over $\eta = \overline{\xi}_0$ we can find specializations $\xi_0 \rightsquigarrow \xi_1 \rightsquigarrow \ldots \rightsquigarrow \xi_d$ lying over $\overline{\xi}_0 \rightsquigarrow \overline{\xi}_1 \rightsquigarrow \ldots \rightsquigarrow \overline{\xi}_d$. In other words we have

$$\xi_{-r} \rightsquigarrow \ldots \rightsquigarrow \xi_{-1} \rightsquigarrow \xi_0 \rightsquigarrow \xi_1 \rightsquigarrow \ldots \rightsquigarrow \xi_d$$

which means that $\dim(X) \geq d + r$ as desired. □